

Vertex Configuration Spaces in 2-Uniform Tessellations

Type 1: (3,4,6,4)

Vertex A: Triangle-Square-Hexagon-Square
Vertex B: Different polygon arrangement
Angles must sum to 360° at each type

Example sum of angles at A: 360°

Type 2: Mixed Triangles & Hexagons

Vertex A: 3-6-3-6
Vertex B: 6-6-6
Both alternating pattern

Vertex A angle sum: 360°

Lattice Basis Types

Square lattice basis:
Two orthogonal vectors

Hexagonal lattice basis:
Two 60° vectors

Key Constraints for k-uniformity

$\forall$ vertex type i :
 $\sum_j (\text{interior angles at } v_i) = 360^\circ$
 $\exists$ precisely k distinct vertex orbits
All polygons are regular
Edge-to-edge property